Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/19/2025
PI Name:	Miller, Fletcher J Ph.D.		
Project Title:	Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	San Diego	**State:**	CA
Zip Code:	92182-1323	**Congressional District:**	53
Comments:	Previously at National Center for Space Exploration Research, NASA Glenn Research Center, until 2007.		
Project Type:		**Solicitation / Funding Source:**	
Start Date:	08/16/2019	**End Date:**	08/15/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	3
No. of Master's Candidates:	7	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/15/2026 per D. Stocker/NASA-GRC (Ed., 9/30/25). NOTE: End date changed to 08/15/2025 per J. McQuillen/NASA-GRC (Ed., 12/23/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, Sandra Ph.D. (NASA Glenn Research Center) Wichman, Indrek Ph.D. (Michigan State University/ grant 80NSSC21M0131)		
Grant/Contract No.:	80NSSC19M0212		
Performance Goal No.:			
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Task Description:	NOTE (Ed., November 2019): Continuation of "Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test," grant #NNX10AD96A, 1/1/2010 to 12/31/2014; and "Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test (NNX14AC70A)," 2/1/2014 to 1/31/2017, with the same Principal Investigator Dr. Fletcher Miller. The research proposed here is directed toward further development of a new apparatus to replace or augment the current NASA Standard 6001 Test 1 used to screen materials for space flight applications, and specifically the flight experiment that will allow us to obtain flame spread data on thick fuel. Test 1 is by far the most widely used test in the Standard, but does not take into account the influence of flow in a controlled manner, while flow has been shown to have a large (often dominant) effect on flame spread in microgravity. The ground-based apparatus described here, The Narrow Channel Apparatus, has been used by the author and collaborators for several years to study fundamental aspects of material flammability in reduced buoyancy conditions. Two new versions of the apparatus have been built that allow the study of flame spread on thick fuels under low pressure, high oxygen conditions such as being considered for new space craft, as well as allow examination of effects of apparatus orientation and sample width. Replacing the Test 1 apparatus with a Narrow Channel will allow for rapid, efficient testing, while also taking into account the low flows actually encountered in microgravity. By controlling the flow, it should be possible to approximate partial gravity conditions; although it is not a major part of this work, the possibility will be explored in the new vertical narrow channel. This phase of the research continues to be devoted to ground-based testing while also preparing for the flight experiments. The work is conducted collaboratively at three institutions by investigators that have worked together for several years in this area (Dr. S. Olson at NASA Glenn Research Center and Dr. I. Wichman at Michigan State University are the other two investigators). The research at San Diego State University consists of five main tasks. The first task will be to make flame spread measurements in a Narrow Channel in low pressure, high oxygen (normoxic) conditions on thick Poly(methyl methacrylate) (PMMA) to match the planned flight test matrix. The second task involves extending our existing Fire Dynamics Simulator (FDS) and/or FLUENT model of flame spread in a Narrow Channel to evaluate the influence of heat loss to the upper wall as well as determine velocity and other field data not easily accessible experimentally. In this part of the research FDS will be run with Gpyro to determine the effect of solid-phase heat conduction. The model will also be run to predict differences between normal, partial, and microgravity including some runs with the fuel surface aligned with gravity. The third task will be to continue to work with NASA to develop and fly a flight experiment to test flame spread with forced flow over a thermally thick fuel in microgravity. In support of this effort, we may do some drop tower tests. There is a large data gap in this subject area, and true microgravity verifications tests are needed to prove the validity of the Narrow Channel Apparatus. After the flight, the data will be analyzed and compared to the model. The fourth task will be to perform Narrow Channel experiments on fuels/solids other than PMMA such as Nomex, Kapton, etc. in a higher oxygen, lower pressure environment. We have a selection of fuels based on discussions with NASA Kennedy Space Center (KSC) and NASA Johnson Space Center (JSC). The last task was added new in 2020 and that was the development of a hybrid Test 1 that is conducted in a vertical orientation in a chamber, but with a low upward flow of the test atmosphere to purge combustion products and allow for a much small apparatus. This will provide comparison data to the Narrow Channel results as well as to tests run previously by NASA in the standard Test 1 apparatus.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research encompasses flame spread and material flammability in narrow channels or gaps. We use this geometry to simulate microgravity conditions. However, many fire situations involve flames in narrow gaps. TWA Flight 800, which crashed off the coast of Canada, was brought down by a small fire spreading through some narrow gaps in the walls and ceiling near the cockpit, for instance. Many building fires involve fire spread inside walls. Fires in computers likewise can occur between the printed circuit boards. Thus, information gained from experiments about how flames spread in various atmospheric conditions in narrow gaps is useful in areas well beyond spacecraft fire safety. The numerical model we are developing will be validated against our experiments and can also be used to study flame spread in these gaps.
Task Progress:	GOALS AND OBJECTIVES The major goals of this project are: • To develop the Narrow Channel Apparatus as a possible supplement or alternative to NASA Standard 6001B Test 1 (material flammability). • To prepare for experiments on the International Space Station (ISS) in 2027 wherein thermally thick fuel slabs would be tested for the first time in long-duration, true microgravity. To accomplish these goals, a number of objectives are being pursued: • Conduct experiments in our existing Narrow Channel Apparatuses (NCA) on thermally thick and thin materials. Vary the oxygen, pressure, and opposed flow velocities. • Measure the fuel surface temperature ahead of the flame to determine the effect of conductive preheating by the flame as it spreads. • Measure the gas-phase flame temperatures for calibrating a numerical model. • Continue developing the numerical model of the flame spread process in the NCA using computational fluid dynamics. • Conduct tests in a modified chamber to produce NASA Test 1 results. • Make measurements in a replica of the flow duct on the ISS to determine the effect of pressure on the fans and the flow distribution. • Develop a new NCA that incorporates lessons learned from the first generation NCAs. ACCOMPLISHMENTS During this reporting period we made progress on all of the objectives listed above. Experiments were conducted on 5 mm thick Poly(methyl methacrylate) (PMMA) in our NCA at oxygen levels from 21% (atmospheric) to 38.8 %, and pressures ranging from 8 psia to 14.7 psia. Flame extinction and flame spread rate was quantified. The optical properties of PMMA were measured in a spectrometer to determine the emissivity so that an infrared camera could be used to measure the PMMA surface temperature during the spreading process; in addition, thermocouples were used for some tests and comparisons. Measurable preheating of over 20 mm was found, showing that for these very slow-moving flames solid-phase conduction can play a role. We also developed a technique for using thermocouples to measure the gas-phase temperature for the first time. Results indicated that the computational model was overestimating the temperature so that we need to make modifications to that. During this period significant progress was made on further developing the numerical model in Ansys Fluent®. Our previous model treated only the gas-phase domain with the window and the fuel surface as boundary conditions. Now

	we have added both the fused silica window and the PMMA fuel to the model so that they play a role in distributing the heat from the flame to environment and control the fuel vapor coming from the PMMA. Since our model is steady state (due to computer limitations), using an experimentally determined spread rate, we added motion to both the window and the fuel to simulate the movement of the flame. This was shown to significantly affect the thermal fields. We also added surface-to-surface radiation to the model to account for fuel radiative losses to the window. Tests were conducted in our NASA Test 1 upward burning chamber. We modified the make-up gas inlet to occupy the entire bottom of the chamber (rather than small side ports) to allow a higher inlet mass flow without increasing the inlet velocity. Tests were run using cotton samples at a variety of pressures and oxygen concentrations since those were the most readily available in the NASA Materials and Processes Technical Information System (MAPTIS) database to which we wanted to compare. We found that our results were more repeatable and consistent than those reported in the MAPTIS database, indicating that our modified Test 1 chamber (not only the NCA) may be of interest to NASA as well. We made additional measurements on our Solid Fuel Ignition and Extinction (SoFIE) flow duct replica using a mass-flow controller as the gas inlet source (rather than the fans that we and SoFIE on the ISS use) and found discrepancies between the fan and the mass flow controller. This could be due to a calibration issue, or due to a flow maldistribution in the duct. To improve the calibration, we designed a new calibration nozzle, using a mix of analytical and numerical techniques, and are in the process of fabricating it. To test for a flow maldistribution, we built a scanning system so we can measure across the entire duct cross-section once the sensor is calibrated.
Bibliography Type:	Description: (Last Updated: 06/15/2021)